

MAT 324 HW10

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Problem 1 (Axler 8A #17). Let λ denote Lebesgue measure on $[1, \infty)$.

(a) Prove that if $f: [1, \infty) \rightarrow [0, \infty)$ is Borel measurable, then

$$\left(\int_{[1, \infty)} f \, d\lambda \right)^2 \leq \int_{[1, \infty)} x^2 f(x)^2 \, d\lambda(x).$$

(b) Describe the set of Borel measurable functions $f: [1, \infty) \rightarrow [0, \infty)$ such that the inequality in (a) is an equality.

deal with the LHS being infinite

Solution. If the left-hand side is ∞ , then so is the right-hand side because...; and if the right-hand side is infinite then the inequality is trivial. Otherwise, f and $x^2 f(x)^2$ are both in the L^2 -space $L^2([1, \infty); \mathbb{R}_{\geq 0})$ of nonnegative square-integrable functions, so by Cauchy-Schwarz on the functions $xf(x)$ and $1/x$,

$$\left(\int_{[1, \infty)} xf(x) \cdot \frac{1}{x} \, d\lambda(x) \right)^2 \leq \left(\int_{[1, \infty)} x^2 f(x)^2 \, d\lambda(x) \right) \underbrace{\left(\int_{[1, \infty)} \frac{1}{x^2} \, d\lambda(x) \right)}_{=1/1-1/\infty=1}. \quad \square$$

Problem 2. Let $(X, \mathcal{S}, \mu)$ be a measure space where there exist disjoint $A, B \in \mathcal{S}$ of positive, finite measure. Show that the norm $\|\bullet\|_p$ on $L^p(\mu)$ is not induced by an inner product on $L^p(\mu)$ for any $p \in [1, \infty] - \{2\}$.

Solution. If $\|\bullet\|_p$ is induced by an inner product, then the parallelogram law holds for the $\mathcal{S}$ -measurable functions $\mathbb{1}_A, \mathbb{1}_B: X \rightarrow \mathbb{R}$, i.e.

$$\|\mathbb{1}_A + \mathbb{1}_B\|_p^2 + \|\mathbb{1}_A - \mathbb{1}_B\|_p^2 = 2\|\mathbb{1}_A\|_p^2 + 2\|\mathbb{1}_B\|_p^2.$$

Since A and B are disjoint, $|\mathbb{1}_A + \mathbb{1}_B| = |\mathbb{1}_A - \mathbb{1}_B| = \mathbb{1}_{A \cup B}$, so the sum and difference of the two indicator functions have the same L^p -norm. Thus, after dividing out a factor of two,

$$(\mu(A) + \mu(B))^{2/p} = \mu(A \cup B)^{2/p} = \mu(A)^{2/p} + \mu(B)^{2/p}. \quad (*)$$

Lemma 2.1. Let $x, y, p \in \mathbb{R}^+$. The equality $x^p + y^p = (x + y)^p$ holds if and only if $p = 1$.

Proof. Let $t = x/y$, so the given equality rewrites as $t^p + 1 = (t + 1)^p$ upon dividing by y^p . Then, we want to show that $f_p(t) = (t + 1)^p - t^p - 1$ has no positive zeros. Note that $f'_p(t) = p(t + 1)^{p-1} - pt^{p-1}$; we analyze the sign of the derivative based on the value of p .

- If $p - 1 > 0$, then $(t + 1)^{p-1} > t^{p-1}$ for all $t \geq 0$, meaning f is increasing on $\mathbb{R}_{\geq 0}$. Since $f_p(0) = 0$, f_p is positive for all positive t .
- If $p - 1 < 0$, then $(t + 1)^{p-1} < t^{p-1}$ for all $t > 0$, meaning f_p is decreasing on $\mathbb{R}^+$. Since f_p is continuous on $\mathbb{R}_{\geq 0}$, it is decreasing on the closure $\mathbb{R}_{\geq 0}$ of $\mathbb{R}^+$, so, since $f_p(0) = 0$, f_p is negative for all positive t .

Thus if $p \neq 1$, the equality doesn't hold, and of course if $p = 1$ then $f_p(t) = 0$ everywhere. $\square$

Applying the lemma with $x = \mu(A)$ and $y = \mu(B)$, we get that $(*)$ holds if and only if $2/p = 1$, hence the parallelogram law doesn't hold if $p \neq 2$, so $\|\bullet\|_p$ is not induced by an inner product for $p \neq 2$. $\square$

Problem 3. Let $C^{0,2}(\mathbb{R}) \subseteq L^2(\mathbb{R})$ denote the subspace of continuous square-integrable functions. Define

$$L_0^2(\mathbb{R}) = \{f \in L^2(\mathbb{R}) : f = 0 \text{ a.e. on } [0, 1]\}, \quad C_0^{0,2}(\mathbb{R}) = L_0^2(\mathbb{R}) \cap C^{0,2}(\mathbb{R}).$$

(a) Let $f \in C^{0,2}(\mathbb{R})$ be such that $f(x) \neq 0$ for some $x \in \mathbb{R} \setminus [0, 1]$. Show that there exists $g \in C_0^{0,2}(\mathbb{R})$ such that $\langle f, g \rangle_2 \neq 0$. Conclude that $f \in C_0^{0,2}(\mathbb{R})$ has a projection to $C_0^{0,2}(\mathbb{R})$ if and only if $f(0) = f(1) = 0$.

(b) Let $f \in L^2(\mathbb{R})$. Determine the projection of f to $L_0^2(\mathbb{R})$.

Solution. (a) Suppose $f(x_0) \neq 0$ for $x \in \mathbb{R} \setminus [0, 1]$; we can assume without loss of generality that $f(x_0) > 0$. Since f is continuous at $x_0 \in \mathbb{R} \setminus [0, 1]$, there is an open interval $I \subseteq \mathbb{R} \setminus [0, 1]$ containing x_0 on which f is positive. Then, let $g : \mathbb{R} \rightarrow [0, 1]$ be a continuous function whose support is contained in I and for which $g(x_0) = 1$. This means $g \in C_0^{0,2}(\mathbb{R})$, $f(x)g(x) = 0$ for $x \in \mathbb{R} \setminus I$, and since $f(x_0)g(x_0) > 0$, $\int_I fg \, d\lambda > 0$, so

$$\langle f, g \rangle_2 = \int_{\mathbb{R}} fg \, d\lambda = \int_I fg \, d\lambda + \underbrace{\int_{\mathbb{R} \setminus I} fg \, d\lambda}_{=0} > 0.$$

Now, we deal with the existence of the projection of f onto $C_0^{0,2}(\mathbb{R})$.

Lemma 3.1. The distance from f to $C_0^{0,2}(\mathbb{R})$ is $\|f|_{[0,1]}\|_2$.

Proof. For $\delta > 0$, let $h_\delta \in C_0^{0,2}(\mathbb{R})$ be defined as

$$h_\delta(x) = \begin{cases} 0 & \text{if } x \in [0, 1] \\ f(x) & \text{if } x \in \mathbb{R} \setminus (-\delta, 1 + \delta) \\ x \cdot \frac{f(-\delta)}{-\delta} & \text{if } x \in (-\delta, 0) \\ (x - 1) \cdot \frac{f(1 + \delta)}{\delta} & \text{if } x \in (1, 1 + \delta). \end{cases}$$

This is easily checked to be a continuous function that agrees with f everywhere except on $(-\delta, 1 + \delta)$. Letting M be the maximum of $|f|$ over $\mathbb{R}$ (which exists because f^2 has finite integral), h_δ is at most $2M$ on the intervals $(-\delta, 0)$ and $(1, 1 + \delta)$, so

$$\|f - h_\delta\|_2^2 = \int_{\mathbb{R}} |f - h_\delta|^2 \, d\lambda \leq \int_{(-\delta, 0) \cup (1, 1 + \delta)} |f - h_\delta|^2 \, d\lambda + \int_{[0, 1]} |f|^2 \, d\lambda \leq (2M)^2 \delta + \|f \cdot \mathbb{1}_{[0,1]}\|_2^2.$$

As δ goes to zero, we get that the distance from f to $C_0^{0,2}(\mathbb{R})$ is bounded above by $\|f|_{[0,1]}\|_2$. Of course, $\|f \cdot \mathbb{1}_{[0,1]}\|_2$ is also a lower bound on the distance because $\|f - h\|_2^2 = \int_{\mathbb{R}} |f - h|^2 \, d\lambda \geq \int_{[0,1]} |f|^2 \, d\lambda$ whenever h is zero on $[0, 1]$; this completes the proof. $\square$

To finish, $h_f \in C_0^{0,2}(\mathbb{R})$ is the projection of f onto $C_0^{0,2}(\mathbb{R})$ if and only if

$$\|f - h_f\|_2^2 = \|f \cdot \mathbb{1}_{[0,1]}\|_2^2 \iff \int_{\mathbb{R} \setminus [0,1]} |f - h_f|^2 \, d\lambda = 0.$$

The latter equality is equivalent to $f - h_f$ being the zero function, which is equivalent to $f - h_f$ being zero on the closure $\mathbb{R} \setminus (0, 1)$ of $\mathbb{R} \setminus [0, 1]$. Since $f - h_f$ and f agree at 0 and 1, such an h_f can only exist if $f(0) = f(1) = 0$, and indeed if $f(0) = f(1) = 0$ then $h_f = f \cdot \mathbb{1}_{[0,1]}$ is a continuous function that minimizes the distance between f and a point of $C_0^{0,2}(\mathbb{R})$.

(b) The distance from $f \in L^2(\mathbb{R})$ to $L_0^2(\mathbb{R})$ can be bounded below by $\|f \cdot \mathbb{1}_{[0,1]}\|_2$ because

$$\|f - h\|_2^2 = \int_{\mathbb{R}} |f - h|^2 \, d\lambda \geq \int_{[0,1]} |f|^2 \, d\lambda$$

whenever h is zero on $[0, 1]$, and the lower bound is achieved by taking h to be $h_f = f \cdot \mathbb{1}_{[0,1]}$, so this h_f is the point minimizing the distance from f to a point of $L_0^2(\mathbb{R})$, which is by definition the projection. $\square$

Problem 4. Let $\mathbb{I} = [0, 1]$ and $L^2(\mathbb{I}; \mathbb{C})$ be the L^2 -space of (equivalence classes of) $\mathbb{C}$ -valued functions on $\mathbb{I}$. For each $n \in \mathbb{Z}$ and $f \in L^2(\mathbb{I}; \mathbb{C})$, define

$$\psi_n: \mathbb{I} \rightarrow \mathbb{C}, \quad \psi_n(x) = e^{2\pi i n x}, \quad c_n(f) = \langle f, \psi_n \rangle_2 = \int_{\mathbb{I}} f \overline{\psi_n} d\lambda \in \mathbb{C}. \quad (1)$$

(a) Show that the collection $\{\psi_n : n \in \mathbb{Z}\}$ consists of orthonormal elements of $L^2(\mathbb{I}; \mathbb{C})$.

(b) Let $f \in L^2(\mathbb{I}; \mathbb{C})$. Show that the sum

$$\sum_{n \in \mathbb{Z}} c_n(f) \psi_n := \lim_{k, m \rightarrow \infty} \sum_{n=-k}^m c_n(f) \psi_n$$

converges with respect to the L^2 norm to an $h_f \in L^2(\mathbb{I}; \mathbb{C})$ so that $\|h_f\|_2 \leq \|f\|_2$ and $\langle f - h_f, \psi_n \rangle_2 = 0$ for all $n \in \mathbb{Z}$.

(c) Suppose f is twice continuously differentiable, $f(0) = f(1)$, and $f'(0) = f'(1)$. Show that for all $n \neq 0$,

$$c_n(f) = \frac{1}{2\pi i n} c_n(f') = -\frac{1}{4\pi^2 n^2} c_n(f'').$$

(d) Under the assumptions on f in (c), show that the sum in (b) converges uniformly to a continuous function $h_f: \mathbb{I} \rightarrow \mathbb{C}$.

Solution. (a) If $m = n$ then $\psi_m(x) \overline{\psi_n(x)} = e^{2\pi i(m-n)x}$ is 1 for all $x \in \mathbb{I}$, so $\langle \psi_m, \psi_m \rangle_2 = 1$ for all $m \in \mathbb{Z}$, and if $m \neq n$ then

$$\langle \psi_m, \psi_n \rangle_2 = \int_{\mathbb{I}} e^{2\pi i(m-n)x} d\lambda(x) = \frac{1}{2\pi i(m-n)} (e^{2\pi i(m-n)} - e^0) = 0.$$

(b) Axler's 8.57 implies that

$$\sum_{n \in \mathbb{Z}} |c_n(f)|^2 = \sum_{n \in \mathbb{Z}} \|c_n(f) \psi_n\|_2^2 \leq \|f\|_2^2,$$

which by Axler's 8.54 implies that $\sum_{n \in \mathbb{Z}} c_n(f) \psi_n$ converges in the L^2 norm to some h_f and

$$\left\| \sum_{n \in \mathbb{Z}} c_n(f) \psi_n \right\|_2^2 = \sum_{n \in \mathbb{Z}} |c_n(f)|^2 \leq \|f\|_2^2.$$

Furthermore, by Axler's 8.58(b), $h_f \in \overline{\text{span}\{\psi_n : n \in \mathbb{Z}\}}$ is equal to $\sum_{n \in \mathbb{Z}} \langle h_f, \psi_n \rangle \psi_n$

$$\langle f - h_f, \psi_n \rangle_2 = \langle f, \psi_n \rangle_2 - \sum_{m \in \mathbb{Z}} \langle c_m(f) \psi_m, \psi_n \rangle_2 = \langle f, \psi_n \rangle_2 - c_n(f) = 0.$$

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(c) For any continuously differentiable function g with $g(0) = g(1)$, integration by parts yields

$$c_n(g) = \int_{\mathbb{I}} g \overline{\psi_n} d\lambda = \frac{1}{2\pi i(-n)} (g(1) \overline{\psi_n(1)} - g(0) \overline{\psi_n(0)}) - \int_{\mathbb{I}} g' \cdot \frac{1}{2\pi i(-n)} \overline{\psi_n} d\lambda = \frac{1}{2\pi i n} c_n(g').$$

Applying this to $g = f$ and $g = f'$ gives us what we want.

(d) For each nonzero integer n ,

$$|c_n(f)| = \frac{1}{4\pi^2 n^2} |c_n(f'')| \leq \frac{1}{4\pi^2 n^2} \int_{\mathbb{I}} |f'' \overline{\psi_n}| d\lambda = \frac{1}{4\pi^2 n^2} \|f''\|_1,$$

so, since $\sum_{n=1}^{\infty} \frac{1}{4\pi^2 n^2} < \infty$,

$$\sum_{n \in \mathbb{Z}} |c_n(f)| \leq |c_0(f)| + 2 \sum_{n=1}^{\infty} \frac{1}{4\pi^2 n^2} \|f''\|_1 < \infty,$$

which implies that the sum in (b) converges uniformly to h_f by the Weierstrass M -test since $|c_n(f)\psi_n| = |c_n(f)|$ for all x . The summands $c_n(f)\psi_n$ are all continuous, so this implies h_f is continuous as well. $\square$

Problem 5. Let $\mathbb{I} = [0, 1]$, $L^2(\mathbb{I}, \mathbb{C})$ be the L^2 -space of $\mathbb{C}$ -valued functions on $\mathbb{I}$, $f \in L^2(\mathbb{I}, \mathbb{C})$, and ψ_n and $c_n(f)$ be as in (1).

- (a) Suppose f is continuous and $f(0) = f(1)$. Show that, given $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ and $a_n \in \mathbb{C}$ for $n = -N, \dots, N-1, N$ such that

$$\left| f(x) - \sum_{n=-N}^N a_n \psi_n(x) \right| \leq \varepsilon \quad \text{for all } x \in \mathbb{I}.$$

- (b) Suppose f is twice continuously differentiable, $f(0) = f(1)$, and $f'(0) = f'(1)$. Show that

$$f = \sum_{n \in \mathbb{Z}} c_n(f) \psi_n$$

with the sum converging uniformly on $\mathbb{I}$.

- (c) Show that the sum in (b) converges to f with respect to the L^2 norm for every $f \in L^2(\mathbb{I}, \mathbb{C})$.

Solution. (a)

(b)

(c)

$\square$